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Labor Supply Responses and Welfare Effects of Tax Reforms*

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Abstract

During the late 1980s and early 1990s, taxation of wage income has undergone a change towards a proportional tax regime in many countries. Our analysis shows that gradual tax reforms in Norway may have removed some of the distortions on worker behavior. Moreover, we find that proportional taxes may reduce the inequality in the distribution of disposable household income. Yet, when the distributional consequences are related to changes in individual welfare, we find that rich households may benefit far more than households at the other tail of the income distribution, because they earn more without any significant increase in effort.

I. Introduction

A common feature of the Scandinavian welfare states, in particular in the first three decades of the postwar period, has been strong political support for a redistributive policy that narrowed income differentials through a progressive income tax. For instance, in 1979 the Norwegian marginal tax rate on incomes in the top decile of the income distribution was at an all-time high and equal to 74 percent. By contrast, the marginal tax rate on incomes in the bottom decile was 32 percent.

The prevailing view among economists in the 1960s and early 1970s was that the distortions on worker behavior caused by a progressive income tax were rather modest. Since the late 1970s there has been a

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growing concern among economists regarding the extent to which high and progressive taxes are eroding the efficiency of the market economy through the disincentives and distortions created by the tax system; see Sandmo (1991) for a historical review of the changing attitudes among economists in the postwar period. Today, public finance scholars, in particular, believe that taxes on labor income cause major distortions on worker behaviour.

In recent years we have witnessed a similar shift in attitudes among leading politicians. Consequently, during the 1980s and early 1990s the marginal tax rates on high labor incomes in Norway were gradually reduced, with the most substantial reduction taking place from 1991 to 1992. The marginal tax rates on incomes in the bottom decile of the income distribution have been changed moderately. As a result, the progressive income tax of today is much less steep than in the late 1970s.¹

One objective of this paper is to examine the distortion on worker behavior caused by the progressive income tax of 1979, that is before the gradual reduction in (formal) marginal tax rates took place. Another objective is to examine the distributional consequences of two tax reforms, one hypothetical and one real. The tax system of 1979 is evaluated against a proportional tax regime and against the tax rules of 1992.

Our analysis shows that when the tax system of 1979 was replaced by the rules of 1992, mean welfare (defined as the mean of households utility levels) increased. However, the order of magnitude of this increase in mean welfare was less than the mean welfare gain obtained by introducing proportional taxation of wage income. The large difference in mean welfare gains is mainly due to the labor supply responses in the lower tail of the household income distribution. The labor supply incentives, in particular for married females, in the lower tail of the income distribution were much less improved by the 1992 reform than by the hypothetical proportional tax reform. Consequently, further reforms of the present tax system towards proportional taxation of wage income may give a substantial increase in mean welfare and thus a substantial reduction in the cost of taxation. Moreover, our analysis shows that such further reforms of the tax system may reduce the inequality in the distribution of after-tax household income. This surprising result is driven by the labor supply

¹ It should be emphasized, however, that there could be a considerable difference between the effective and formal marginal tax rates. In 1979 all interest payments were deductible against a maximal marginal tax rate of 74 percent, while in 1992 deductions were only allowed against a marginal tax rate of 28 percent (equal to the marginal tax rate on capital income). These changes in deduction rules implied that although the differences between the formal tax rates of the 1970s and early 1990s were large, the differences between the effective rates were much smaller, in particular for net borrowers in the high income brackets.

responses caused by a proportional tax reform, since we find that this reform induces the “poor” to work much harder than before. For these poor households, the decrease in the value of leisure nearly outweighs the welfare gain related to increased consumption. For the “rich” households, the impact of the proportional tax reform on leisure is rather modest, while consumption increases considerably. Thus, when the distributional consequences are judged in terms of the changes in individual welfare in a heterogeneous population, the rich may benefit far more from further reforms towards proportional taxation of wage income than the poor.

The paper is organized as follows. In the analysis of tax reforms we apply a microeconomic labor supply model. Since microeconomic studies of tax reforms, in which observed as well as unobserved heterogeneity are accounted for, are rather novel, we focus on some methodological issues in Section II. Section III gives a brief outline of the labor supply model. In Section IV we present the empirical specification of the model and the estimation results. The model is estimated on Norwegian labor market and tax return data from 1979 for married couples. Section V gives the policy simulation results, and Section VI summarizes the main findings.

II. Methodological Issues

Our point of departure, on which the policy evaluation methodology is based, is a particular framework of modeling labor supply behavior developed in Dagsvik (1994) and adopted by Dagsvik and Strøm (1995) for the purpose of analyzing labor supply. The modeling approach differs from the conventional empirical models of labor supply, in which labor supply is derived from utility maximization where only consumption and leisure are the choice variables of the household. In these conventional models, agents are supposed to have no preferences over other important job characteristics, such as the nature of the tasks to be performed, location of workplace, social environment and working conditions. The framework proposed in Dagsvik and Strøm (1995) acknowledges these qualitative aspects of jobs by modelling labor supply behavior as a discrete choice problem, where the alternatives are “job packages”. These job packages are characterized by specific attributes such as hours of work, wage rates and other non-pecuniary variables. In addition, this framework is able to take into account that there are important quantity constraints in the market, in the sense that different types of jobs are not equally available to every agent. Agents differ by qualification, and jobs differ with respect to qualification required.

A labor supply model can be used to identify the gains and losses from tax reforms by applying different Hicks-compensating measures; see

Auerbach (1985), Hausman (1984) and King (1987) for a discussion of alternative money metrics of welfare change, and Kay (1980) for arguments in favor of using equivalent variation (EV). In this study we apply EV. Loosely speaking, EV is measured as the amount of money that has to be added to the household's disposable income under the initial tax rules in order to make the household indifferent between the initial and the alternative tax system. Note that EV is measured at the household level. EV sums up the household net welfare gain/loss associated with behavioral responses induced by tax reforms, say, increased consumption and reduced leisure.

In the empirical measurement of the cost of taxation — or of the gain/loss of tax reforms — several approaches have been pursued in the literature. One common approach is to analyze tax reforms on the basis of a computable general equilibrium model with one — or a limited number of — representative household(s) representing labor supply and commodity demand in the economy. A standard reference by now is Ballard *et al.* (1985), and for an application to the Norwegian economy, see Vennemo (1992). The advantage of a general equilibrium approach is that it allows for wage rates and prices to respond to changes in tax rates. One serious objection to the standard approach is that it ignores all — or a substantial part — of the heterogeneity in the economy. Heterogeneous preferences are important not only for estimation, but also for simulation and welfare analysis. This latter aspect has been neglected relative to the attention that heterogeneity has received in the literature on estimation and testing.

Some of the problems that arise in the empirical analysis of tax reforms stem from the existence of non-linearities in the agents' budget constraints. In the representative agent setting in general equilibrium models, important non-linearities in the budget constraints are not accounted for. From a theoretical point of view the presence of non-linearities in the budget constraints implies that there is no representative consumer; see King (1987).

An alternative approach pursued in the literature is to estimate the overall gain from tax reforms as the area under the Hicks-compensated aggregated labor supply curve. In this approach the wage rate is exogenously given, which implies a partial equilibrium setting. As with the general equilibrium modeling strategy, heterogeneity is ignored and no distributional consequences can be traced. A standard reference on this topic is Browning (1987). It should be noted that evaluating tax reforms by aggregating — explicitly or implicitly — welfare gains and losses across households is critical, since it assumes a utilitarian welfare function for society.

In most of the previous microeconomic analyses of labor supply responses and welfare effects of tax reforms, rather simple functional

forms (typically linear and loglinear) were chosen, which tend to make the analysis unconvincing, since the results often do not seem to be robust as regards the choice of functional form; see e.g. Hausman (1985), Blomquist (1983) and Blundell *et al.* (1988).

An empirical micro-model — such as the one we apply here — is designed to account for observed as well as unobserved heterogeneity. Unobserved heterogeneity arises from the fact that as econometricians we are unable to observe all factors that affect individual tastes and opportunities. These unobservables are modeled as random variables, i.e., money metrics of welfare change, say EV, at the household level becomes a random variable; see King (1987) and Atkinson (1990). In other words, micro-econometric models allow the analyst to study the distribution of EV which accounts for observed and unobserved heterogeneity in tastes and choice constraints. There has not been a strong tradition in applied welfare analysis to include stochastic terms when a model is applied in policy assessments, as is evident from the prominent works of Hausman (1985), Hausman and Poterba (1987) and Blomquist (1983).

There is no general agreement on how the information captured by EV should be summarized when EV is random. Haneman (1982) discusses two alternative approaches. The first is to use the mean of the distribution as a measure of the overall gain. The second is to apply the median of the distribution. Here, we follow the former approach. In addition, we provide information about other parameters of the distribution of welfare gains and losses across households.

III. A Brief Outline of the Labor Supply Model

For expository simplicity, we begin by focusing on single-person households which, to the analyst, have observationally identical characteristics and constraints.

In our setting each agent faces a different set of non-market and market opportunities (jobs). It is understood that “jobs” may be specified in such detail that there exists a wide range of hours-wage combinations in the market. Let $B_i(h, w)$ denote the set of jobs with hours $h > 0$, and wage rate $w > 0$, that are feasible to agent i . $B_i(0, 0)$ is the set of non-market opportunities. Let $U_i(C, h, j)$ denote the utility for agent i of consumption C , hours h and opportunity j , where $j \in B_i(h, w)$, $h \geq 0$, $w \geq 0$.

The economic budget constraint is given by

$$C = f(wh, I), \quad (1)$$

where I is non-labor income and f is a function that transforms gross income into after-tax income. The price index of the composite good

(called consumption) is equal to one. When inserting the budget constraint into the utility function we get $U_i(f(wh, I), h, j)$.

Assume furthermore that

$$U_i(f(wh, I), h, j) = v(f(wh, I), h) \varepsilon_{ij}(h, w), \quad (2)$$

where $v(C, h)$ is a positive, deterministic function which is quasi-concave in (C, h) , increasing in the first argument and decreasing in the second. The term $\varepsilon_{ij}(h, w)$ is a random taste shifter that is supposed to capture the effect of unobservable attributes associated with opportunity j . Note that this term is viewed as random only from the econometrician's point of view, while it is assumed known to the agent. Specifically, $\{\varepsilon_{ij}(h, w)\}$ accounts for the fact that for a given agent, tastes may vary over opportunities, hours and wages, and for a given opportunity, tastes may vary across agents. We now define

$$V_i(h, w) = \max_{j \in B_i(h, w)} U_i(f(wh, I), h, j). \quad (3)$$

$V_i(h, w)$ is the conditional indirect utility function, given hours of work and the wage rate. In other words, for agent i , $V_i(h, w)$ is the utility of the most preferred opportunity among the feasible opportunities with hours h and wage rate w .

From (2) and (3) we get

$$V_i(h, w) = \psi(h, w) e_i(h, w) \quad (4)$$

where

$$e_i(h, w) = \max_{j \in B_i(h, w)} \varepsilon_{ij}(h, w) \quad (5)$$

and

$$\psi(h, w) = v(f(wh, I), h). \quad (6)$$

Recall that hours and wage rates are fixed for each job so that when a job has been chosen, then hours and wage rate follow. The individual agent is assumed to choose the job that maximizes utility. The corresponding hours and wage rate, (h, w) , therefore follow from maximizing $V_i(h, w)$.

Next we briefly consider the structure of the choice probabilities of realized hours and wage rates, i.e., the hours and wage rate that correspond to the agent's choice. These probabilities enter the likelihood function which is used when estimating the unknown parameters of the model.

Let $n_i(h, w)$ be the number of jobs in $B_i(h, w)$ and let $\bar{g}(h, w)$ be the mean of

$$\frac{n_i(h, w)}{\sum_{x>0} \sum_{y>0} n_i(x, y) + n_i(0, 0)}$$

across agents. Assume that the taste shifters are i.i.d. with distribution

$$P(\varepsilon_{ij}(h, w) \leq y) = \exp\left(-\frac{1}{y}\right). \quad (7)$$

Equation (7) follows from the assumption that agents have preferences over job types that satisfy the “independence from irrelevant alternatives” property. The taste shifters are also assumed to be stochastically independent of the choice sets $\{B_i(h, w)\}$.

Let $\varphi(h, w)$ be the probability that agent i will choose a job with hours and wage rate (h, w) . If the mean of $\sum_{x,y} n_i(x, y)$ is large, the assumptions introduced above imply that

$$\begin{aligned} \varphi(h, w) &\equiv P[V_i(h, w) = \max_{x,y} V_i(x, y)] \\ &= \frac{\psi(h, w) \tilde{g}(h, w)}{\psi(0, 0) \tilde{g}(0, 0) + \sum_{x>0} \sum_{y>0} \psi(x, y) \tilde{g}(x, y)}, \end{aligned} \quad (8)$$

for $h > 0$, $w > 0$. For the sake of interpretation it is convenient to express (8) as

$$\varphi(h, w) = \frac{\psi(h, w) g_0 g(h, w)}{\psi(0, 0) + g_0 \sum_{x>0} \sum_{y>0} \psi(x, y) g(x, y)} \quad (9)$$

where

$$g(h, w) = \frac{\tilde{g}(h, w)}{\sum_{x>0} \sum_{y>0} \tilde{g}(x, y)} \quad (10)$$

and

$$g_0 = \frac{\sum_{x>0} \sum_{y>0} \tilde{g}(x, y)}{\tilde{g}(0, 0)}. \quad (11)$$

The function $g(h, w)$ is denoted the conditional opportunity density of hours and wage rates, and it can be interpreted as the mean of the fraction of feasible jobs with hours h and wage rate w . Similarly, g_0 is the mean of the fraction of feasible opportunities that are job opportunities. Note that

g_0 is not necessarily less than one. The probability of not working equals

$$\varphi(0,0) = \frac{\psi(0,0)g_0}{\psi(0,0) + g_0 \sum_{x>0} \sum_{y>0} \psi(x,y)g(x,y)} \quad (12)$$

The functional form of (9) and (12) is particularly attractive. The labor supply density $\varphi(h, w)$ is expressed as a simple function of the structural term of the utility function, $\psi(\cdot)$, and of $g_0g(\cdot)$, which is an aggregate representation of the set of feasible job opportunities.

The identification of $g_0g(\cdot)$ and $\psi(\cdot)$ and their inter-relationship are discussed in Dagsvik and Strøm (1995); it is found that the identification problem can only be fully solved when additional assumptions about the distribution of offered hours are made. The model discussed here is related to models developed in the travel demand literature; see Ben-Akiva and Lerman (1985).

The extension of the model outlined above to deal with the joint decisions of husband and wife is straightforward. The household utility function is now assumed to have the structure:

$$U_i(C, h_F, h_M, j_F, j_M) = v(C, h_F, h_M) \varepsilon_{ij_Fj_M}(h_F, w_F, h_M, w_F), \quad (13)$$

where F, M index female and male, respectively. Apart from the bivariate indexation of job opportunities, the probabilities of realized hours and wage rates are completely similar to (9) and (12). Below we only show the probability of realized hours and wage rates when both spouses work:

$$\varphi(h_F, h_M, w_F, w_M) = \frac{\psi(h_F, h_M, w_F, w_M) g_{0FM} g_F(h_F, w_F) g_M(h_M, w_M)}{D} \quad (14)$$

where

$$\begin{aligned} D = & g_{0FM} \sum_{x_1 > 0} \sum_{x_2 > 0} \sum_{y_1 > 0} \sum_{y_2 > 0} \psi(x_1, x_2, y_1, y_2) g_F(x_1, y_1) g_M(x_2, y_2) \\ & + g_{0M} \sum_{x_2 > 0} \sum_{y_2 > 0} \psi(0, x_2, 0, y_2) g_M(x_2, y_2) \\ & + g_{0F} \sum_{x_1 > 0} \sum_{y_1 > 0} \psi(x_1, 0, y_1, 0) g_F(x_1, y_1) + \psi(0, 0, 0, 0), \end{aligned} \quad (15)$$

and where $\psi(h_F, h_M, w_F, w_M) = v(f(h_F w_F + h_M w_M, I), h_F, h_M)$, and $f(\cdot)$ represents the budget constraint. The different terms of the opportunity density may be interpreted as follows: $g_k(h_k, w_k)$ is the mean fraction of

jobs with hours and wage rate (h_k, w_k) that are feasible to gender k , $k = F, M$; g_{0FM} is the mean fraction of opportunities that are feasible job opportunities to the couple; g_{0k} is the mean fraction of opportunities that are feasible job opportunities to gender k . Note that the structure of g_{0FM} allows opportunity sets for husband and wife to be correlated. The wife's and the husband's opportunity sets may be correlated because they both live in the same place and consequently operate on the same local labor market.²

IV. Empirical Specification and Estimation Results

In order to estimate an empirical model it remains to specify the structural part of the utility function and the opportunity densities. Let A_k , $k = M, F$ denote the age of spouse k , and let $CU6$ and $CO6$ denote the number of children less than and above six years of age. The deterministic part of the utility function is assumed to have the following functional form:

$$\begin{aligned} \log v(C, h_M, h_F) = & \alpha_2 \frac{(10^{-4}C)^{\alpha_1} - 1}{\alpha_1} \\ & + \left(\frac{L_M^{\alpha_3} - 1}{\alpha_3} \right) (\alpha_4 + \alpha_5 \log A_M + \alpha_6 (\log A_M)^2) \\ & + \left(\frac{L_F^{\alpha_7} - 1}{\alpha_7} \right) (\alpha_8 + \alpha_9 \log A_F + \alpha_{10} (\log A_F)^2) \\ & + \alpha_{11} CU6 + \alpha_{12} CO6 + \alpha_{13} L_F L_M \end{aligned} \quad (16)$$

where $L_k = 1 - h_k/8760$, $k = M, F$. $\alpha_1, \alpha_2, \dots, \alpha_{13}$, are unknown parameters. Note that since 8760 is the total number of annual hours, L_k is equal to normalized annual leisure for gender k . If $\alpha_1 = \alpha_3 = \alpha_7 = 0$, then the utility function is loglinear in consumption and leisure, and if they are equal to one it is linear; for a recent use of this functional form in econometric work, see McFadden and Leonard (1993). If $\alpha_1 < 1$, $\alpha_3 < 1$, $\alpha_7 < 1$, $\alpha_2 > 0$,

$$\alpha_4 + \alpha_5 \log A_M + \alpha_6 (\log A_M)^2 > 0$$

² This model is formulated in a discrete choice setting. In Dagsvik (1994) a pure-choice-of-attribute modeling framework with continuous attributes was developed, from which an empirical labor supply model for married couples was specified and estimated by Dagsvik and Strøm (1995). Thus the model given by (14) and (15) is the discrete version of this model.

and

$$\alpha_8 + \alpha_9 \log A_F + \alpha_{10}(\log A_F)^2 + \alpha_{11} CU6 + \alpha_{12} CO6 > 0,$$

then $v(C, h_M, h_F)$ is increasing in C , decreasing in (h_M, h_F) , and quasi-concave in (C, h_M, h_F) . Note that the marginal utility of income only depends on household consumption and that it is declining with household consumption.

Household consumption is assumed to be equal to disposable income. The f -function that represents the budget constraint follows from the tax and benefit rules. Claimed deductions are also accounted for, but these are suppressed in the notation here. A non-convex budget set may follow when effective marginal taxes are not uniformly increasing with income and/or when spouses are jointly taxed for incomes in certain income brackets. Certain benefit rules may add to this non-convexity of the budget set. Taxable income is equal to gross income minus deductions, which may vary across households. In 1979 parts of the taxes were levied on gross income and other parts on taxable income (description of the tax rules as of 1979 and 1992 are available on request).

We assume that offered hours and wages are independently distributed. It is important to note that this assumption does not exclude that realized hours and wages are dependent. Realized hours and wages follow from the optimal job chosen and this choice depends on preferences. The justification for the assumption that offered hours of work and wages are independent is that in the Scandinavian welfare states, working hours are typically regulated by the government or through central agreements between the organizations in the labor market, independently of wage formation.

In the model specification, this feature is captured by specifying the opportunity density of hours for the female, $g_{1F}(\cdot)$, as

$$\log g_{1F}(h_F) = d_F + \alpha_{19} D_1(h_F) + \alpha_{20} D_2(h_F) \quad (17)$$

where $D_1(h_F)$ is equal to one for full-time hours and zero otherwise, while $D_2(h_F)$ equals one for part-time hours and zero otherwise. For males the corresponding specification equals

$$\log g_{1M}(h_M) = d_M + \alpha_{18} D_1(h_M). \quad (18)$$

The parameters d_F and d_M are determined such that the respective opportunity densities sum up to unity.

We note that (17) and (18) imply that the opportunity densities of hours are assumed to be uniform apart from peaks at full-time for men and at part-time and full-time for women. Unless this or analogous assumptions are made, it is not possible to separate some of the structural coefficients in the systematic part of the utility function from the parameters of the

opportunity densities of hours offered. Note that the case with uniformly distributed offered hours corresponds to the traditional textbook framework where agents are free to choose the desired hours of work. As we argue in this paper, a more realistic assumption is that there are more available part-time and full-time hour jobs than jobs with other working hours.

The offered wage opportunity densities are assumed to be the densities of a lognormal distribution function, with expectations:

$$E \log W_k = \beta_0 + \beta_1 s_k + \beta_2 \text{Exp}_k + \beta_3 (\text{Exp}_k)^2 \times 10^{-2}; \quad k = M, F, \quad (19)$$

where s_k denotes years of schooling for gender k , Exp_k is experience of gender k , equal to $A_k - s_k - 6$.

Estimation Results

The estimation of the model is based on a procedure suggested by McFadden (1978) which yields results that are close to the full maximum likelihood method; for further details see Dagsvik and Strøm (1995). Note that the opportunity densities are jointly estimated with the parameters of the utility function.

The estimates are reported in Table 1 (utility function and opportunity densities related to hours and job opportunities) and Table 2 (wage opportunity density). The estimates imply that the deterministic part of the utility function is an increasing and quasi-concave function in consumption and leisure.

Now, consider the estimates of the opportunity densities. Note first that g_{0FM}/g_{0M} is to be interpreted as the mean fraction of feasible job opportunities for the married couple, relative to the mean fraction of job opportunities for the male. If the males' and females' opportunity sets are independent, we get $g_{0FM}/g_{0M} = g_{0F}$ and $g_{0FM}/g_{0F} = g_{0M}$. For this to be the case we must have that the interaction term $\alpha_{17} = 0$. According to Table 1, this is not true. Furthermore, we see that the females' market opportunities increase by education. We have not found a similar effect for males and we only report the estimation results when this variable is not included in the specification of the male opportunity density of hours.

Figures 1–3, which display the observed and predicted distributions of hours of work and consumption in 1979, demonstrate that the model reproduces the observed distributions quite well.

Labor Supply Elasticities

In labor supply studies it is common to report elasticities computed from mean sample values, or alternatively based on subsamples of households that are grouped according to some socio-demographic characteristics.

Table 1. *Estimates of the parameters of the utility function and of the opportunity density of hours**

Variables	Coefficients	Estimates 1979
<i>Preferences:</i>		
Consumption	α_1	0.916 (22.4)
	α_2	1.997 (9.7)
Male leisure	α_3	-4.782 (6.7)
	α_4	76.657 (2.4)
	α_5	-38.532 (2.3)
	α_6	5.159 (2.3)
Female leisure	α_7	-0.961 (2.8)
	α_8	258.717 (2.7)
	α_9	-136.624 (2.7)
	α_{10}	19.208 (2.8)
	α_{11}	6.573 (8.1)
	α_{12}	1.967 (5.5)
Leisure interaction term	α_{13}	-3.869 (0.8)
<i>Opportunities:</i>		
Female; $\log(g_{0FM}/g_{0M}) = \alpha_{14} + \alpha_{15}S_F$	α_{14}	-1.164 (1.6)
	α_{15}	0.208 (3.0)
Male; $\log(g_{0FM}/g_{0F}) = \alpha_{16}$	α_{16}	-2.414 (7.4)
<i>Interaction</i>		
Female and male; $\log(g_{0FM}/g_{0F}g_{0M}) = \alpha_{17}$	α_{17}	-1.841 (4.6)
<i>Opportunity densities of hours:</i>		
Full-time peak, males	α_{18}	0.765 (5.8)
Full-time peak, females	α_{19}	0.868 (4.2)
Part-time peak, females	α_{20}	0.407 (3.0)

*Numerical *t*-values in parentheses.

Table 2. *Wage opportunity densities; simultaneous ML estimates**

Variables	Males	Females
Intercept	2.711 (39.6)	2.730 (3.52)
Education	0.045 (13.3)	0.047 (9.8)
Experience ($\text{Exp } A_k - s_k - 6$)	0.022 (6.4)	0.008 (2.1)
$\text{Exp}^2 \times 10^{-2}$	-0.036 (-6.2)	-0.012 (-1.7)
Standard error	0.101 (36.0)	0.172 (33.6)
R^2	0.23	0.23

*Numerical *t*-values in parentheses.

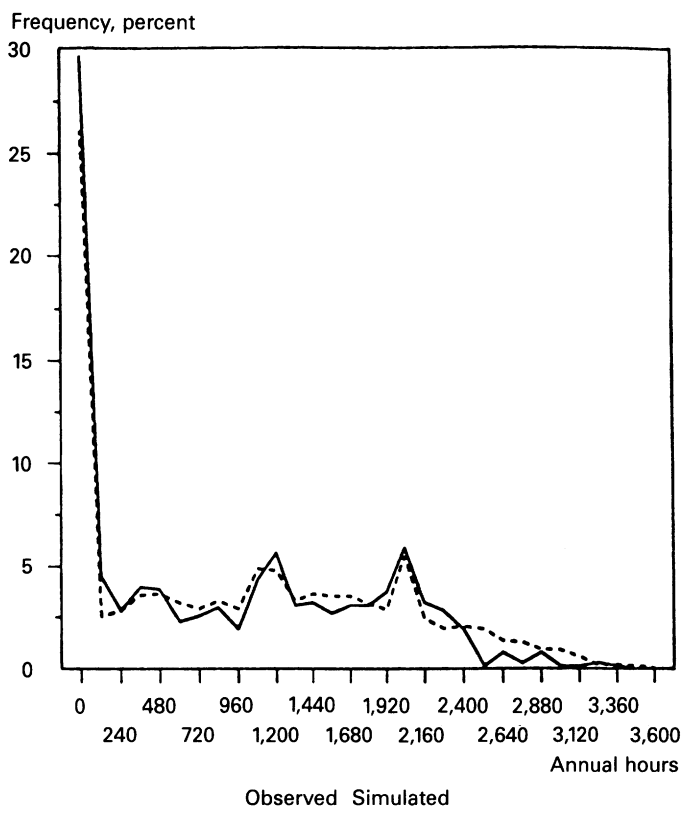


Fig. 1. Observed and simulated hours of work for females.

This is meaningful only when the taste shifters are independent of hours and wages. This is not the case in our model since the taste shifters vary across jobs, and hence across hours and wages; cf. (5). In this model, an increase in hours supplied, say as a response to a higher wage level, will imply a shift of job. Thus we had to take the complete random utility model into account when deriving the elasticities. Hence, we instead computed elasticities of hours of work with respect to the mean wage rate. These elasticities are denoted exact aggregate labor supply elasticities. Note that observed and unobserved heterogeneity in the population as well as the non-convexities of the budget sets, and hours regulations were taken into account when these aggregate elasticities were calculated.

In Table 3 we report aggregate elasticities. The table is divided into three parts. The first part reports the elasticity of the probability of participation with respect to wage, the second the elasticities of hours

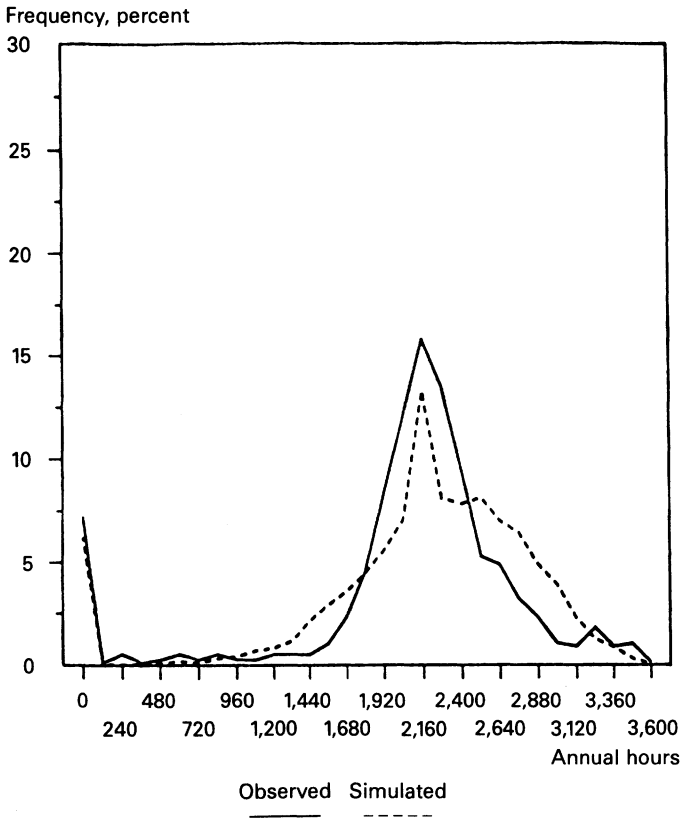


Fig. 2. Observed and simulated hours of work for males.

worked, conditional on participation, while the third reports the elasticities of the unconditional supply of labor. In each part we give elasticities for the 10 percent poorest and richest households, for the 80 percent in the middle and for all households.

Table 3 shows that the elasticities are sharply declining with income. The elasticities of participation tend to be higher than the elasticities of hours supplied, given participation. The elasticities among females are significantly higher than among males, in particular for individuals with middle and higher income.

Increasing the overall wage level by 1 percent yields a direct unconditional Slutsky elasticity of 0.33 for males and of 1.59 for females. When the cross-effects are taken into account, the utility-constant labor supply elasticity is reduced to 0.22 for males and to 1.31 for females.

Most researchers report uncompensated labor supply elasticities that are rather low compared to those we have found, in particular for females.

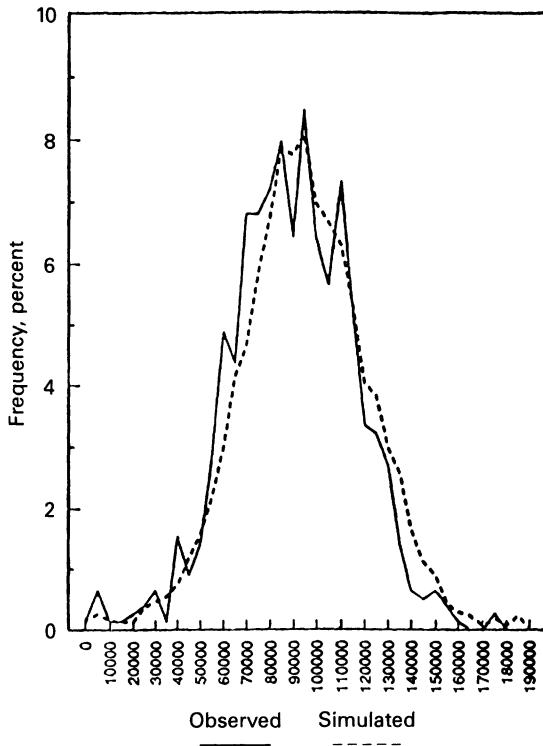


Fig. 3. Observed and simulated consumption for married couples, Norway, 1979.

For example, Blomquist and Hansson-Brusewitz (1990) found labor supply elasticities (hours with respect to wage level) for married women (Sweden) between 0.4 and 0.8. However, exceptions are Arrufat and Zabalza (1986) and Hartog and Theeuwes (1986). Arrufat and Zabalza found an overall elasticity of hours with respect to wage level for married women (U.K.) equal to 2.03, while the elasticity of participation with respect to wage level was equal to 1.41. The corresponding estimates found by Hartog and Theeuwes (the Netherlands) are 3.61 and 1.88. Moreover, Hausman and Ruud (1984) report an elasticity for wives of 0.76. Our male labor supply elasticities are more in line with what others have found; e.g. Blundell and Walker (1986), Meghir (1985) and Pencavel (1987).

V. Policy Simulations

Our methodology for evaluating policy changes is based on EV derived from individual indirect random utilities by means of the model discussed

Table 3. *Exact aggregate labor supply elasticities* for males and females, 1979*

Type of elasticity	Male elasticities		Female elasticities			
	Own wage elasticities	Cross elasticities	Own wage elasticities	Cross elasticities		
Elasticity of the probability of participation	Cournot	I	1.89 (0.06)	-1.04 (0.05)	1.85 (0.07)	-1.44 (0.07)
		II	0.09 (0.003)	-0.08 (0.004)	0.66 (0.008)	-0.29 (0.005)
		III	0.03 (0.004)	0.01 (0.003)	0.07 (0.006)	-0.03 (0.011)
		IV	0.29 (0.004)	-0.08 (0.004)	0.83 (0.009)	-0.25 (0.008)
	Slutsky	I	2.71 (0.35)	0.41 (0.09)	2.62 (0.28)	0.21 (0.21)
		II	0.07 (0.01)	-0.12 (0.01)	0.73 (0.03)	-0.19 (0.02)
		III	0.01 (0.01)	-0.05 (0.02)	0.03 (0.03)	-0.11 (0.03)
		IV	0.22 (0.02)	-0.08 (0.02)	0.73 (0.03)	-0.16 (0.01)
	Cournot	I	0.29 (0.02)	-0.15 (0.02)	1.04 (0.04)	-1.04 (0.07)
		II	0.07 (0.002)	-0.09 (0.004)	0.78 (0.009)	-0.29 (0.006)
		III	0.03 (0.005)	-0.01 (0.005)	0.12 (0.013)	-0.06 (0.017)
		IV	0.16 (0.002)	-0.07 (0.004)	0.99 (0.007)	-0.26 (0.008)
Slutsky	I	1.11 (0.12)	0.47 (0.11)	2.39 (0.24)	0.42 (0.24)	
	II	0.09 (0.02)	-0.05 (0.01)	0.97 (0.05)	-0.17 (0.01)	
	III	0.01 (0.01)	-0.02 (0.01)	0.05 (0.01)	-0.04 (0.02)	
	IV	0.10 (0.01)	-0.02 (0.01)	0.80 (0.04)	-0.12 (0.01)	
Elasticity of unconditional expectation of total supply of hours	Cournot	I	2.23 (0.06)	-1.18 (0.05)	3.09 (0.08)	-2.33 (0.11)
		II	0.16 (0.003)	-0.17 (0.004)	1.49 (0.012)	-0.57 (0.009)
		III	0.06 (0.006)	-0.01 (0.004)	0.19 (0.013)	-0.08 (0.017)
		IV	0.45 (0.005)	-0.15 (0.005)	1.82 (0.11)	-0.51 (0.014)
	Slutsky	I	4.15 (0.49)	0.88 (0.15)	5.68 (0.58)	0.68 (0.45)
		II	0.16 (0.02)	-0.17 (0.02)	1.77 (0.07)	-0.35 (0.03)
		III	0.02 (0.02)	-0.07 (0.02)	0.07 (0.03)	-0.15 (0.05)
		IV	0.33 (0.02)	-0.11 (0.02)	1.59 (0.05)	-0.28 (0.02)

Note that I = 10 percent poorest household, II = 80 percent in the middle of the consumption distribution, III = 10 percent richest household and IV = all households.
 * Standard deviations in parentheses.

above. For expository reasons we focus here on single individuals. The extension to the case of married couples is straightforward. For single individuals it follows (cf. Section II) that the indirect utility has the structure

$$\begin{aligned}\tilde{V}_i(y, f) &= \max_{h, w} (\max_{j \in B_i(h, w)} U_i(f(wh, I) + y, h, j)) \\ &= \max_{h, w} (v(f(wh, I) + y, h) e_i(h, w))\end{aligned}\quad (20)$$

where $e_i(h, w)$ is defined in (5). Moreover, this random term can be expressed as $e_i(h, w) = g_0 g(h, w) \theta_i(h, w)$ when $h, w > 0$, and $e_i(0, 0) = \theta_i(0, 0)$ when $h = w = 0$, where $\{\theta_i(x, y)\}$ are i.i.d. with cumulative distribution function

$$P(\theta_i(h, w) \leq x) = \exp\left(-\frac{1}{x}\right). \quad (21)$$

Hence we can write (20) as

$$\begin{aligned}\tilde{V}_i(y, f) &= \\ \max_{h > 0, w > 0} [&\max (v(f(wh, I) + y, h) g_0 g(h, w) \theta_i(h, w)), v(f(0, I) + y, 0) \theta_i(0, 0)].\end{aligned}\quad (22)$$

In words, (20) means that first, utility is maximized with respect to feasible market and non-market opportunities with hours h and wage rate w , subject to the budget constraint

$$C = f(wh, I) + y \quad (23)$$

where y is a given non-taxable amount of money. This gives the conditional indirect utility given (h, w) ; i.e., the highest utility attainable for the agent, given that hours of work and the wage rate are equal to (h, w) , or given that the agent does not work. Second, this conditional indirect utility is maximized with respect to (h, w) which yields the unconditional indirect utility as a function of non-labor income, the tax system, represented by f , and non-taxable income y . This indirect utility also depends on the opportunity density, $g_0 g(\cdot)$, that represents the choice sets $\{B_i(h, w)\}$.

By means of the formalism above, the definition of EV can be expressed as

$$\tilde{V}_i(0, f_1) = \tilde{V}_i(\text{EV}, f_0) \quad (24)$$

where f_1 represents the alternative tax system and f_0 represents the initial system.

The policy experiments are performed in a partial equilibrium setting, as in Browning (1987). This means that the opportunity density, $g_0 g(\cdot)$, is kept fixed. In other words, the distribution of offered hours and wage rates

associated with feasible jobs is assumed to remain unaffected by changes in the tax rates. For example, if total labor supply increases as a result of tax reforms, then the total number of jobs is assumed to increase so as to maintain a fixed number of job opportunities for each worker.

Since the random terms $\{\theta_i(h, w)\}$ are i.i.d. with a distribution function that does not depend on the structural part of the model, it is easy to use (22) and (24) to perform simulations. In the simulation experiments reported below we generated values of the random terms $\{\theta_i(h, w)\}$ by means of (21).

Consider now the design of the Norwegian personal income tax systems under consideration. The base situation — or the pre-reform tax system — is the tax system as of 1979. The following reforms were investigated: (i) proportional taxation of wage income, (ii) the 1992 tax system and (iii) lump-sum taxes.

The motivation for analyzing these regimes may be explained as follows (see also Section I). First, marginal tax rates as of 1979 were at an all-time high level. Second, marginal tax rates were gradually cut throughout the 1980s and the beginning of the 1990s; the most substantial cut took place in 1992. Third, the slashing of tax rates has gradually moved the tax system towards a proportional tax system for wage income. Fourth, it is not fully satisfactory to use no-taxation or lump-sum taxes as a benchmark case since the latter system cannot be implemented in practice. However, as shown below, the supply-side potential is not much lower with proportional taxes than with lump-sum taxes.

When proportional taxation of wage income replaces the 1979 rules, the tax rate is chosen so as to yield a tax revenue equal to the tax revenue under the 1979 regime. Given this constraint, we simulated the mean gain (mean of EV) relative to mean tax revenue. Moreover, we report additional information on the distribution of individual welfare gains and losses. The extent to which the reform affects the distribution of gross and disposable income is also examined. In doing this we account for both behavioral and non-behavioral effects. Note that in conventional analyses of tax reforms carried out by central authorities, behavioral responses are typically disregarded.

When the 1979 tax system is replaced by the 1992 system, revenues obviously cannot remain unchanged. Thus, we report the mean of the gain from this reform relative to the change in the mean tax revenue.

Proportional Taxes

We applied the model to determine the proportional tax rate which ensures that tax revenue under proportional taxation remains equal to tax revenue under the 1979 system. This rate was found to be 20.1 percent.

Table 4. *Labor supply,* tax and income under three different tax regimes; means in 1979 NOK*

	Participation rates percent		Annual hours of work		Household		
	F	M	F	M	Gross income	Taxes	Disposable income
1979 rules	74.2	93.7	1,043	2,132	136,279	40,623	95,656
Proportional taxes	84.3	99.5	1,642	2,790	208,449	40,623	167,825
1992 rules	74.5	96.0	1,178	2,331	156,720	48,809	107,911

*Supply of hours, given participation, is equal to annual hours divided by the participation rates.

Table 4 reports the participation rates and annual hours supplied by gender under the different tax regimes. The table also lists gross household income and disposable income.

The results show that a shift from the 1979 tax rules to proportional taxation of wage income: (i) increases participation rates for females and males by approximately 10 and 6 percentage points, respectively; (ii) increases annual hours supplied, given participation, for females and males by 38 percent and 31 percent, respectively; and (iii) increases gross and disposable household income by 53 percent and 75 percent, respectively. The mean of EV is found to be equal to NOK 25 956 (1979 prices), which amounts to 63.8 percent of the tax revenue under the 1979 rules.

Tables 5a and 5b provide information on the distribution of the gains and losses. When 1979 rules are replaced by proportional taxes, then: (i) as much as 99.4 percent of the population will gain from this reform, while 0.6 percent will lose; (ii) inequality in the distribution of gains proves to be considerably higher than the inequality in the distribution of gross income in 1979; (iii) among the 10 percent of the population that gains the most from the reform, the mean of EV is NOK 45 400 and, for these households, average gross income under the 1979 rules was NOK 182 800 (in contrast, the equivalent amounts for the 10 percent that gains the least are NOK 1 600 and NOK 84 500).

The latter result demonstrates that the magnitude of the gains from the proportional tax reform increases strongly with pre-reform gross household income. The explanation is that among the “poor”, the tax reform induces the individuals to work much longer hours, as indicated by the reported elasticities in Table 3, and the gain for these households occurs as a result of increased income and hence consumption. Among the “rich”, hours of work are not affected to the same extent. On the other

Table 5A. *Equivalent variation, 1979 NOK*

	Equivalent variations				
	Losers		Winners		
	Total	Percent of population	Mean	Gini coefficient	Percent of population
Proportional taxes 1992 rules	Mean				
	25,956	0.6	-1,344	—	99.4
	1,588	42.8	-2,356	0.437	56.8
					Mean
					26,136
					4,572
					Gini coefficient
					0.303
					0.472

Table 5B. *Mean EV and gross household income of the 10 percent who gain most and the 10 percent who gain least from a proportional tax reform, 1979 NOK*

	Mean equivalent variation	Gross household income under 1979 rules
The 10 percent who gain most	45,400	182,800
The 10 percent who gain least (lose most)	1,600	84,500

hand, the “rich” benefit greatly from the direct increase in disposable income.

Table 6 reports the Gini coefficients of the distributions of gross and disposable household income. These results demonstrate that the inequality in the distribution of disposable income decreases when proportional taxes replace the progressive 1979 system. This surprising result is due to the fact that the poor have higher labor supply elasticities than the rich. The tax reform induces the poor to choose jobs with much longer hours of work than under the 1979 tax rules.

Even though income inequality decreases due to this reform, the welfare implications, as measured by EV, turn out to be more favorable for the medium-rich and rich households than for the lower-middle and poor households.

The 1992 Tax System

We realize from Table 4 that the 1992 tax reform has a more modest effect on labor supply than proportional taxation. Note, however, that the assumption of a fixed tax revenue is relaxed. The revenue based on the 1992 system turns out to rise by approximately 20 percent which, on average, corresponds to NOK 8186 per household. Despite an increased tax burden, the transition from 1979 to 1992 tax rules yields a positive mean equivalent variation. As reported in Table 4, the mean equivalent variation is equal to NOK 1588 which appears to be considerably lower than the increase in tax revenue. The ratio between the mean equivalent variation and the mean tax increase is equal to 0.194, which turns out to be in accordance with results for the US reported by Browning (1987).

Tables 5A and B show that there is large heterogeneity with regard to gains and losses from this tax reform. As much as 43 percent lose from the reform while the remaining 57 percent are winners. Table 6 shows that the inequality in gross household and disposable income increases.

Table 6. *Inequality in the distributions of gross and disposable income measured by the Gini coefficient**

	Gross income	Disposable income
1979 tax rules	0.196 (0.002)	0.161 (0.002)
Proportional taxes	0.152 (0.001)	0.152 (0.001)
1992 tax rules	0.205 (0.003)	0.177 (0.003)

*Standard deviations in parentheses.

Lump Sum Taxes

In studies of the cost of taxation, lump-sum taxes often serves as the benchmark case with which alternative tax systems are compared. The rationale for doing this is that lump-sum taxes remove all distortive effects of taxation and reveal the supply-side potential of the economy. Since lump-sum taxes cannot be implemented in the economy, it is important to consider alternative reforms (to lump-sum taxes). However, the specific reforms should be evaluated both against the base system of 1979 and against lump-sum taxes in order to reveal the labor supply potential. As shown in Table 7, female labor supply under the 1979 tax rules constitutes 48.2 percent of the non-distortive potential, while in the case of males the figure is 72.2 percent. When proportional taxes replace the 1979 tax system, utilization of the labor supply potential increases considerably for both sexes. In the case of male labor supply it is close to full utilization of its potential, while for females the reform brings utilization up to 76 percent of its potential. The 1992 reform turns out to increase male labor supply to some extent, but much less than the proportional tax system, while female labor supply is closer to the outcome under the 1979 rules. Table 7 also shows how far gross income is from its supply-side potential under the alternative tax systems. Based on the 1979 tax rules, less than 60 percent of the potential turns out to be utilized, while the 1992 reform has increased utilization up to around two-thirds of its potential. We observe, however, that proportional taxes would have brought the economy up to 88 percent of the gross income potential.

VI. Conclusions

The measurement of the welfare effects of tax reforms discussed in this paper is based on a particular empirical labor supply model developed and estimated in Dagsvik and Strøm (1995). In this model labor supply

Table 7. Mean annual hours of work and gross household income under alternative tax regimes in percent of the corresponding lump-sum results

	Annual hours of work		Gross household income
	F	M	
1979 tax rules	48.2	72.2	57.7
Proportional taxation	76.0	94.5	88.3
1992 tax rules	54.5	78.9	66.4
Lump-sum taxes			
Hours, NOK 1979	2,160	2,952	236,197

decisions are viewed as choices between feasible “job packages”, where each package is characterized by a wage rate, hours of work and unobserved non-pecuniary attributes. This labor supply model allows for non-convex budget sets, joint decisions of husband and wife, regulations of hours of work, and observed as well as unobserved heterogeneity in preferences and opportunity sets.

In line with others, e.g. Browning (1987), this study applies a partial-equilibrium approach in which the distribution of feasible market and non-market opportunities is kept fixed under various tax-reform experiments. Thus, we are tacitly assuming that supply creates its own demand. In contrast to most studies of the cost of taxation, this study accounts for the effects of observed as well as unobserved heterogeneity when the different reforms are evaluated.

In recent years the degree of progression in the taxation of wage income in Norway has gradually been reduced. Our analysis shows that these gradual reforms may have removed some of the distortion on worker behavior caused by income taxes. However, further reforms towards a proportional tax regime may increase mean welfare and improve efficiency much more. In fact, our results indicate that a proportional tax regime may move the economy close to its supply-side potential as revealed by lump-sum taxation.

The impact on the distribution of welfare gains and losses across households indicates that a proportional tax reform will favor households with incomes between the middle and the top decile of the income distribution far more than the households in the bottom decile. The reason is that the Hicks-compensated labor supply elasticities are much higher among the low-income households than among the high-income households. The proportional tax reform induces the poor to work much more than before and the decreased utility associated with decreased leisure nearly outweighs the benefit from increased disposable income. For the rich, and typically well-educated, the supply elasticities are much lower; they therefore enjoy the increase in income without any large decrease in the utility generated by less leisure.

As regards income distribution, the effect of replacing the 1979 rules by proportional taxes is a reduction in the inequality of disposable household income. However, this surprising result is not inconsistent with the reported distribution of welfare gains. As alluded to above, proportional taxation of wage income induces the poor to work so much harder that the inequality in the distribution of gross income as well as in the distribution of disposable income decreases.

Nearly all households (99.4 percent) gain from replacing the progressive tax system of 1979 by proportional taxation, given a fixed tax revenue. Thus, switching to proportional taxation almost implies a Pareto improve-

ment. The mean welfare gain of the 1979 regime — evaluated against proportional taxation — turns out to be equal to 64 percent of tax revenue. However, our analysis demonstrates that the gains vary strongly with the level of pre-reform 1979 incomes. For instance, if we use a social welfare function that exclusively adds up the equivalent variation of the 10 percent of the households that gain the least, then the mean welfare gain is approximately 4 percent of initial tax revenue. Alternatively, when the social welfare function is based on the equivalent variation of the 10 percent that gain the most, the mean welfare gain increases to approximately 119 percent. These results illustrate the importance of accounting for heterogeneity when evaluating the welfare effects of tax reforms.

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